Fiscal Year:	FY 2023	**Task Last Updated:**	FY 07/13/2023
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Project Title:	Cold-Sleep for Long-duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 BRASH2101: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	06/01/2022	**End Date:**	10/31/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2024 per TRISH Science Office (Ed., 3/14/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Buysse, Daniel M.D. (University of Pittsburgh) Empey, Philip Ph.D. (University of Pittsburgh) Guyette, Francis Ph.D. (University of Pittsburgh) Weissman, Alexandra M.D. (University of Pittsburgh)		
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Task Description:	We propose a protocol for a subset of the crew to adopt a temporary low-resource status, which we call “cold-sleep,” to mitigate psychological stress and reduce resource utilization. Cold-sleep will reduce psychological stress by decreasing perceived duration of mission segments where crewmembers do not have specified tasks. Cold-sleep will reduce oxygen and food consumption as well as carbon dioxide and solid waste production. Cold-sleep uses drugs to increase the normal ratio of sleeping:wakefulness time for a crewmember from ~6 hours:18 hours to ~20 hours:4 hours with a mild 1-2ºC reduction in core body temperature during sleep. This target should reduce perceived mission duration by over 75% and metabolic requirements by 50%. In prior human experiments by our group, we found that the intravenous (IV) alpha-2-adrenergic receptor (A2AR) agonist dexmedetomidine induces a light sleep, mild hypothermia, and reduces resting oxygen consumption (VO2) by 25-30% relative to quiet resting. In a series of laboratory protocols in healthy human volunteers, we will solve several scientific, design, and technical challenges to bring A2AR-drug induced cold-sleep to a complete working prototype. Aim 1: Demonstrate similar metabolic suppression with oral, topical, or transmucosal A2AR agonists. This step is essential to eliminate the need for long-term IV access and to allow self-administration by a crewmember. Aim 2: Demonstrate a comfortable and reliable monitoring array and cooling pad to ensure astronaut safety and temperature control. This step will translate the invasive techniques we find expedient in short-term laboratory experiments to a practical solution that will be palatable and easily used in long-term missions. Aim 3: Demonstrate the actual metabolic savings, perceived time, and rate of recovery to normal function after a short and medium length cold-sleep regimen. This will demonstrate readiness of the cold-sleep protocol. Our protocols will establish the pathway for testing the efficacy of other emerging drugs for metabolic suppression that are not yet approved in humans.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Original Aims / Objectives: We will solve several technical challenges to bring alpha-2-adrenergic receptor (A2AR)-drug induced cold-sleep from countermeasure readiness level (CRL) 5 (components demonstrated in laboratory) to CRL 6 (complete working prototype tested in laboratory). We will demonstrate reducing waking time by over 75% (18 hours/day to 4 hours/day) and total energy requirements by 30% (~35 kcal/kg/day to ~25 kcal/kg/day). Our protocols will establish the pathway for testing the clinical efficacy of other emerging approaches for resource conservation or metabolic manipulation. Aim 1: Select an oral or transmucosal A2AR agonist to induce cold-sleep. This step is essential to eliminate the need for intravenous access and to allow self-administration by a crew member. We will design dosing schedules by measuring duration of effects. Aim 2: Validate non-invasive sensors for monitoring and controlling temperature during cold-sleep. This step will translate invasive techniques used in short-term laboratory experiments to a solution that is usable and feasible for long-term missions as well as determining optimal placement, size and settings for cooling pads. Aim 3: Measure reduction in energy expenditure and rate of recovery after 1-day or 5-day cold-sleep. This will demonstrate the readiness of the cold-sleep protocol (CRL 6). This step will also examine the subjective experience of cold-sleep and any carryover effects of cold-sleep after recovery. Key Findings: We have screened 50 participants, who have completed 39 protocol days in the laboratory to advance Aim 1. During each protocol, we made primary measurements of (1) energy expenditure using indirect calorimetry and (2) core body temperature (ingested telemetry capsules), as well as multiple other physiological measures. Specifically, we tested whether single oral doses of dexmedetomidine (1 µg/kg sublingual or 4 µg/kg swallowed) or tizanidine (8 mg or 16 mg), affect energy expenditure and core body temperature in humans (n=8-9 participants/dose). Dexmedetomidine induced a median 13% to 19% peak reduction and tizanidine induced a median 15% to 22% peak reduction in energy expenditure relative to baseline. After dexmedetomidine and tizanidine, core body temperature decreased a median of 0.41ºC to 0.64 ºC and 0.43ºC to 0.63ºC respectively. Decreases in temperature occurred after peak reductions in metabolic rate and were associated with increased energy expenditure for participants after 1µg/kg dexmedetomidine and 8 mg tizanidine, but not after 4 µg/kg dexmedetomidine or 16 mg tizanidine. Plasma drug levels were higher after the higher dose of each drug, and dexmedetomidine levels were related to mean temperature change. Other physiological effects were inconsistently related to plasma drug levels. We conclude that both oral dexmedetomidine and tizanidine directly reduce energy expenditure prior to changing temperature and the higher doses of these drugs can suppress cold-induced thermogenesis in a manner that allows induction of mild hypothermia in humans. Based on these results, we have tested sustained administration of dexmedetomidine (2 µg/kg oral dose followed by 1µg/kg/hr subcutaneous administration for 6 hours) in 4 participants (of 8 planned). In these participants, temperature decreases by 2ºC and energy expenditure decreases 20%-25%. Subjective effects of the drugs last 6-8 hours. We conclude that this drug strategy can deliver the physiological changes we sought. We plan to complete this series of experiments, measure plasma drug levels, and select the final dose of drug to use for Aim 3. For Aim 2, we have tested an established telemetry capsule (HQI). This system often had loss of connectivity because other laboratory equipment would interfere with the radio signal. This resulted in missing or artifactual data. We obtained a newer telemetry capsule (GreenTeq). This capsule has an internal memory and Bluetooth connections that have essentially eliminated data loss. We will use this capsule moving forward. In addition, we have placed multiple surface temperature monitors on participants during their protocol days. Two commercial products include a dual-sensor device (Calera Core Temperature Sensor) to estimate core temperature, and the Bair Spot-On, which uses isothermal skin measurements to estimate core temperature. The Calera device has moderate correlation with core temperature. The Spot-On has good correlation with core temperature. Both are perturbed if air currents blow directly on the sensor. We have also measured heat flux from different skin regions. These measurements may be better indicators of core body temperature change. For Aim 3, we have obtained Institutional Review Board (IRB) approval to study metabolic rate and activity in a series of ambulatory participants who are not given any drug. These experiments will provide normative data on energy expenditure, monitoring and sleep. We will begin this experiment in the next month. We anticipate beginning 1-day sleep studies in late summer or early fall.

	Impact of Findings: Data so far confirm that we can induce a torpor like state with lower energy expenditure in humans for at least 6-8 hours. We have identified non-invasive devices that can monitor for the safety of participants. We have enough data to select a dose of dexmedetomidine for Aim 3. Proposed Research Plan: (1) Complete subcutaneous dexmedetomidine series - 5 more participant days. (2) Complete ambulatory measurements of energy expenditure - 10-15 participants. (3) Complete 1-day protocol - 8-16 participant days. (4) Complete 5-day protocol - 8 participant days. Impact: 1. This project has developed new knowledge about torpor in humans: a. Dexmedetomidine and tizanidine inhibit Cold-Induced Thermogenesis (CIT). Both of these alpha-2-adrenergic drugs prevented the expected rise in energy expenditure as core body temperature decreased. Both were most effective at the higher doses. b. Dexmedetomidine and tizanidine reduce energy expenditure (EE). This decrease is coincident with the sedating effects of the drugs. c. Drug-induced reduction of EE occurs prior to fall in core body temperature. This observation is consistent with data from hibernating animals, but slightly different than our original concept for this project. While hypothermia does reduce basal metabolic rate, the temperature-dependent reduction may not be as large as the primary effect of sedation. d. Hypothermia reduces EE. We analyzed our prior work with intravenous dexmedetomidine, and we determined that VO2 decreases ~ 5.6% / ºC as long as shivering (cold-induced thermogenesis) is inhibited. This corresponds to a Q10 of 2.8. These are new data for humans. 2. This project is advancing art of human temperature measurement: a. Available surface monitors can estimate core body temperature. This will be useful in medical settings or in settings occupational settings where exposure might lead to hypothermia. b. Total heat flux of the body and energy expenditure can model heat balance. This provides insight into regulation of core body temperature, and we are defining the measures that will be important for future studies of temperature manipulation in humans. c. We demonstrated that similar heat flux is afforded by different temperatures of water in our cooling device (Arctic Sun). This has allowed us to accomplish core body temperature reduction with mild water temperatures (15ºC) that are less noxious to the skin. 3. We have identified utility of alpha-2-adrenergic drugs as a sedative or hypnotic: a. These drugs may have different side-effects than other sedatives (e.g. less ataxia and memory impairment than benzodiazepines). b. These drugs may be formulated in oral or topical forms for medical use
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Articles in Peer-reviewed Journals	Flickinger KL, Weissman A, Elmer J, Coppler PJ, Guyette FX, Repine MJ, Dezfulian C, Hopkins D, Frisch A, Doshi AA, Rittenberger JC, Callaway CW. "Metabolic manipulation and therapeutic hypothermia." Ther Hypothermia Temp Manag. 2023 Jul 3. https://doi.org/10.1089/ther.2023.0010 ; PMID: 37405749 , Jul-2023